MIL-C-48243D (AR)
28 March 1986
SUPERSEDING
MIL-C-48243C (AR)
1 February 1979

MILITARY SPECIFICATION

CHARGE PROPELLING, 155MM, M203, LOADING, ASSEMBLING AND PACKING

This Specification is approved for use within the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing for one type of propelling charge designated as Charge Propelling, M203, for 155MM Howitzer, M198.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and Supplement thereto, listed in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-223	-	Powder, Black
MIL-C-43157	-	Cloth, Spun Viscose Rayon Resin Impregnated
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1320

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-414	-	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection By Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

9281897 -	Charge, Propelling, M203 for 155MM Howitzer, M198
9293303	Packing and Marking for Container, Ammo, Metal: PA68 for Charge, Propelling, 155MM, M203 for Howitzer, M198

(Copies of specifications, standards, handbooks, drawings, and publications and other Government documents required by suppliers in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials - Materials shall be in accordance with applicable drawings and specifications.

3.2. Propelling charge. - The propelling charge shall comply with all requirements specified in applicable drawings and specifications.

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3.3 Moisture content of black powder. - The moisture content of the black powder at the loading station at the time of loading shall not exceed the amount specified in MIL-P-223.

3.4 Weights

3.4.1 Propellant weight. - The propellant weight shall comply with the weight specified in the loading authorization within plus or minus 0.25 ounce.

3.4.2 Charge, propelling assembly weight. - The charge, propelling assembly weight shall not differ by more than 0.75 ounce from the total amount specified in the loading authorization plus actual weight of the empty increment and complete igniter assembly plus the average or actual weight of the flash reducer and average weight of the lacing cord and lacing jacket (see 6.8).

3.4.3 Complete igniter assembly weight - (see 6.8).

3.4.3.1 Actual weight of tube. - The complete igniter assembly weight shall not differ by more than 0.15 ounce from the nominal black powder weight plus the actual weight of the tube and cap, plus the actual or average weight of the empty base igniter assembly and the body and strap assembly.

3.4.3.2 Average weight of tube. - The complete igniter assembly weight shall not differ by more than 0.30 ounce from the nominal black powder weight plus the average tube and cap weight plus the actual or average weight of the empty base igniter assembly and the body and strap assembly

3.5 Dimensional gages

3.5.1 Maximum length and diameter. - The completed charge shall freely enter the prescribed gage. The gage shall be designed to inspect the maximum diameter and length. If the diameter feature is separate from the length feature, the diameter feature must extend approximately the full length of the charge.

3.5.2 Minimum diameter. - The base end of the completed charge shall not freely enter the prescribed gage greater than two inches.

3.6 Proving ground. (See 6.15) The muzzle velocity, pressure, and velocity standard deviation shall be as specified in Table I.

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TABLE I Velocity and pressure requirements.

Zone	Proj	Temp	Velocity (feet per second) Minimum	Maximum	Velocity Standard Deviation FPS	Individual Maximum Pressure* PSI
Chg. 8	M549	70°F	2675	2745	7.0	50,000
Chg. 8	M549	145°F	(Differential pressure)		-	57,950
Chg. 8	M549	-65°F	(Differential pressure)		-	NA

*See 6.16

Note: The PXR-6325 projectile simulator may be used as an alternate to the inert loaded M549 RA projectile.

3.6.1 Differential pressure

3.6.1.1 Hot firings. No individual maximum differential pressure shall exceed 6000 psi, or if the mean plus three (3) times the standard deviation exceeds 6000 psi, the lot will be referred.

3.6.1.2 Cold firings. - No individual differential pressure shall exceed 3000 psi or if the mean plus three (3) times the standard deviation exceeds 3000 psi, the lot will be referred (see 6.14). In addition, if any individual ignition delay exceeds 150 milliseconds or if the mean plus three (3) times the standard deviation exceeds 150 milliseconds, the lot will be referred (see 6.14).

3.7 Assembly torque. - The test plug shall be assembled to withstand re-application of the minimum assembly torque specified on the applicable drawing, without moving (see 6.18)

3.8 Igniter tubes. All tubes shall be free of cracks, dents, creases, tears, flat areas and softness greater than 0.25 inch.

3.9 First article inspection. This specification contains technical provisions for the first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.10 Complete igniter alignment. An approved method of centering the complete igniter assembly in the increment assembly will be utilized for all charges.

3.11 Workmanship. All components shall be free of dirt, grease and other foreign material. There shall be no discoloration or deterioration of cloth and no evidence of rips, tears or other cloth defects. All work performed shall be performed in a thorough workmanlike manner.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. - The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection. - (See 6.11)

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. This sample shall consist of the following:

Part Description	Drawing	Quantity
Strap, Tying	9278988	5
Igniter Core Bag Assembly	9281905	5
End, Base Igniter	9278990	5
Base Igniter End and Igniter Core Bag Assembly	9278986	5
Base Igniter Assembly	9278985	5
Base Igniter Loading Assembly	9281936	5
Complete Igniter Assembly	9281940	5
Liner Assembly	9278971	5
Disc, Flash Reducer	9289201	5
Flash Reducer Loading Assembly	9289200	5
Increment Loading Assembly	9281939	5
Charge Propelling Assembly	9281938	5
Jacket Lacing	9281895	5
Charge, Propelling, M203 for 155MM Howitzer M198 (in sealed containers)	9281897/ 9293303	10
Cloth, Spun Viscose Rayon, Resin Impregnated, Class 2		4 square yards
Cloth, Spun Viscose Rayon, Resin Impregnated, Class 3		4 square yards

4.3.2 Inspection to be performed. - The First Article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to any or all the requirements of the applicable drawings and specifications and to the inspections of Table II. All inspections listed in Table II shall be classified Major unless otherwise specified in referenced paragraphs and specifications.

4.3.3 Rejection. - See MIL-A-48078

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TABLE II First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
	Propelling Charge Components and Assemblies				
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Strap Tying (Dwg. 9278988) Examination for Defects	5		3.2	4.4.2.2
	Igniter Core Bag Assembly (Dwg. 9281905) Examination for Defects	5		3.2	4.4.2.4
	End Base Igniter (Dwg. 9278990) Examination for Defects	5		3.2	4.4.2.5
	Base Igniter End and Igniter Core Bag Assembly (Dwg. 9278986) Examination for Defects	5		3.2	4.4.2.6
	Base Igniter Assembly (Dwg. 9278985) Examination for Defects	5		3.2	4.4.2.7
	Base Igniter Loading Assembly (Dwg 9281936) Examination for Defects (see note a)	5		3.2	4.4.2.9

NOTES:

a. All defects except major defect 102

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TABLE II - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48243D(AR)

PARAGRAPH	TITLE Continuation Sheet		SHEET 2 OF 3		DRAWING NUMBER NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Complete Igniter Assembly (Dwg. 9281940) Examination for Defects (Note b)	5		3.2	4.4.2.11
	Liner Assembly (Dwg 9278971) Examination for Defects (Note b)	5		3.2	4.4.2.13
	Disc, Flash Reducer (Dwg. 9289201) Examination for Defects	5		3.2	4.4.2.18
	Flash Reducer Loading Assembly (Dwg 9289200) Examination for Defects	5		3.2	4.4.2.19
	Increment Loading Assembly (Dwg. 9281939) Examination for Defects	5		3.2	4.4.2.20
	Jacket, Lacing (Dwg, 9281895) Examination for Defects	5		3.2	4.4.2.22

NOTES: b. All defects except Major 102

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TABLE II First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48243D(AR)

PARAGRAPH	TITLE		3 SHEET OF 3		DRAWING NUMBER
	Continuation Sheet				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Charge Propelling, M203 for 155MM Howitzer M198 (in sealed container) (Dwg. 9281897 and 9293303) Examination for defects (Note c)	10			4.4.2.23/4.4.2.24/ 4.4.2.25/4.4.2.26/ 4.4.2.27
	Cloth, Spun Viscose Rayon, Resin Impregnated, Class 2	Note d		Note e	Note f
	Cloth, Spun Viscose Rayon, Resin Impregnated, Class 3	Note d		Note e	Note f

Notes:
c. All defects except: Critical 2, para 4.4.2.23; Special defect a., para 4.4.2.23; Major defects 101, 105 and 108, paragraph 4.4.2.23; Major defects 102 and 104, para 4.4.2.25.
d. See Military Specification MIL-C-43157.
e. All physical characteristics listed in Table I in para 3.5 of Military Specification MIL-C-43157.
f. See para 4.3.1 of Military Specification MIL-C-43157.

NOTES:

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of propelling charge assemblies shall contain:

a. Propellant from not more than one lot number from one manufacturer.

b. Black powder from not more than one lot number from one maufacturer.

c. Tubes from not more than one interfix number from one manufacturer.

4.4.2 Examination. - See MIL-A-48078.

a. Sampling plans. - Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Body	1 OF 1	9278987
			NEXT HIGHER ASSEMBLY 9278984

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Cloth improper			3.1	4.5.5.2
Minor 201	Width (as established)	Note a	0.65%	3.2	Scale

NOTES: a/ If a slitting machine is utilized, in lieu of the specified AQL, during the slitting process each roll will be inspected within the first 10 feet and the last 10 feet; i.e. two measurements per roll.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.2	Strap, Tying				9278988
					NEXT HIGHER ASSEMBLY 9278984
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Cloth improper			3.1	4.5.5.2
Minor 201	Length		0.65%	3.2	Scale
202	Width	note a	0.65%	3.2	Scale

NOTES: a/ If a slitting machine is utilized, in lieu of the specified AQL, during the splitting process each roll will be inspected within the first 10 feet and the last 10 feet; i.e., two measurements per roll.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.3	Bag, Igniter Core		9281904
			NEXT HIGHER ASSEMBLY 9281905

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Width of cloth	Note a	0.40%	3.2	Scale
102	Cloth improper			3.1	4.5.5.1
Minor					
201	Evidence of poor workmanship		0.65%	3.11	visual

NOTES: a/ If a slitting machine is utilized, in lieu of the specified AQL, during the slitting process each rool will be inspected within the first 10 feet and the last 10 feet; i.e., two measurements per roll.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.4	Igniter Core Bag Assembly	1	1	9281905
				NEXT HIGHER ASSEMBLY 9278986

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Tie missing or improperly located		100%	3.2	Gage
102	Location of seams		100%	3.2	Visual
103	Cloth improper (tape)			3.1	4.5.5.2
Minor					
201	Length of tape after tack		0.65%	3.2	Scale
202	Length of tape before tack		0.65%	3.2	Scale
203	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.5	End, Base Igniter	1	1	9278990
				NEXT HIGHER ASSEMBLY: 9278985/9278986

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cloth improper			3.1	4.5.5.1
102	Diameter of base igniter end		0.40%	3.2	Gage
Minor					
201	Marking missing, misleading, or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.6	Base Igniter End and Igniter Core Bag Assembly	1	1	9278986
				NEXT HIGHER ASSEMBLY 9278985

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Distance from end base igniter to first tape		100%	3.2	Gage
102	Location of igniter core bag assembly		100%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

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PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.7	Base Igniter Assembly		9278985
			NEXT HIGHER ASSEMBLY 9281936

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Diameter of seam circle		0.40%	3.2	Scale
102	Quilting missing or improper (including length)		0.40%	3.2	Visual-Manual
Minor 201	Evidence of poor workmanship		0.65%	3.11	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48243D (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.8	Body and Strap Assembly				9278984
					NEXT HIGHER ASSEMBLY 9278983
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Tying strap missing, incompletely stitched or improperly located		0.40%	3.2	Visual-Scale
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Base Igniter Loading Assembly				9281936
					NEXT HIGHER ASSEMBLY 9278983
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Assembly incorrect or damaged to the extent that black powder can escape (including any seams or openings incompletely stitched or broken threads)		100%	3.2	Visual-Manual
Major 101	Weight of black powder		100%	3.2	4.5.1
102	Moisture content of black powder		a/	3.3	4.5.3
103	Distance from end of igniter core to seam		0.40%	3.2	Scale
104	Any seam incorrect, or incompletely stitched		100%	3.2	Visual
Minor 201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

a/ See paragraph 4.4.3.2

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.10	Cap (Alternate: Tube Closure)	1	1	9289199

NEXT HIGHER ASSEMBLY: 9281940/9345159

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Thickness		0.40%	3.2	Gage
Minor 201	Length		0.65%	3.2	Scale

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
4.4.2.11	Complete Igniter Assembly		1	1	9281940
					NEXT HIGHER ASSEMBLY 9281939
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Assembly incorrect or damaged to the extent that black powder can escape (including any seams or openings incompletely stitched or broken stitches)		100%	3.2	Visual
2	Igniter core twisted or folded in tube (to be checked prior to application of cap or tube closure)		100%	3.2	Visual
Major					
101	Base igniter loading assembly improperly tied to tube; i.e., tape missing, loose, or incorrect (excluding length of tape after tying)		100%	3.2	Visual-Manual
102	Complete igniter assembly weight		100%	3.4.3	4.5.9
103	Cap or tube closure missing or insecure		100%	3.2	Visual-Manual
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.12	Body	1	1	9278958
				NEXT HIGHER ASSEMBLY 9278973

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Cloth improper			3.1	4.5.5.2
Minor 201	Length (as established)		0.65%	3.2	Scale
202	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9278971
4.4.2.13	Liner Assembly				NEXT HIGHER ASSEMBLY 9278973
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Lead foil not properly matched or not properly assembled to the scrim side of the additive liner		0.40%	3.2	Visual
102	Weight of lead foil prior to laminating to scrim		0.40%	3.2	Balance
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9278973
4.4.2.14	Body and Liner Assembly				NEXT HIGHER ASSEMBLY 9278967
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Liner assembly improper, improperly assembled, missing or improperly located		0.40%	3.2	Visual-Scale
102	Minimum distance from seams to edge of cloth		0.40%	3.2	Visual
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF		DRAWING NUMBER 9278977
4.4.2.15	End, Increment				NEXT HIGHER ASSEMBLY 9278974
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Cloth improper			3.1	4.5.5.2
Minor 201	Established length		0.65%	3.2	Scale
202	Established width of straight section		0.65%	3.2	Scale

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48243D (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.16	Increment End Assembly	1	1	9278974
				NEXT HIGHER ASSEMBLY 9278967

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Seam improper		0.40%	3.2	Visual-Manual
Minor 201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9278967
4.4.2.17	Body and End Assembly				NEXT HIGHER ASSEMBLY 9281939
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	End assembly improperly assembled to body and liner assembly		0.40%	3.2	Visual
102	Any seam incompletely stitched		0.40%	3.2	Visual-Manual
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48243D (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.18	Disc, Flash Reducer	1	1	9289201
				NEXT HIGHER ASSEMBLY: 9289216

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Cloth improper			3.1	4.5.5.2
Minor 201	Diameter of flash reducer disc		0.65%	3.2	Scale
202	Marking missing, misleading, or unidentifiable		0.65%	3.2	Visual
203	Inner diameter of disc		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.19	Flash Reducer Assembly	1	1	9289216
				NEXT HIGHER ASSEMBLY: 9289200

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Seam incompletely stitched		0.40%	3.2	Visual
Minor 201	Evidence of poor workmsnship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48243D (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.20	Flash Reducer Loading Assembly	1	1	9289200
				NEXT HIGHER ASSEMBLY 9281938

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Potassium sulfate weight after loading		0.40%	3.2	Balance
102	Assembly damaged or improper to the extent potassium sulfate can escape		100%	3.2	Visual
Minor 201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.21	Increment Loading Assembly				9281939
					NEXT HIGHER ASSEMBLY
					9281938
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Propellant weight		100%	3.4.1	4.5.2
102	Complete igniter assembly improper or damaged (including condition of tube immediately prior to insertion into body and end assembly)		100%	3.2/3.8	Visual-Manual
103	Complete igniter assembly improperly tacked to body and end assembly		100%	3.2	Visual-Manual
104	Increment damaged to the extent that propellant can escape (includes any seams or openings incompletely stitched and broken threads)		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.22	Jacket, Lacing	1	1	9281895
				NEXT HIGHER ASSEMBLY 9281897

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cloth improper			3.1	4.5.5.2
102	Lacing cord missing or improperly assembled (2 locations)		0.40%	3.2	Visual-Manual
Minor					
201	Jacket of excessive length		0.65%	3.2	Scale
202	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
203	Any seam improper or incompletely stitched		0.65%	3.2	visual-Manual
204	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.23	Charge, Propelling, M203 for 155MM Howitzer M198	1	3	9291897
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Assembly damaged or improper to the extent that black powder can escape		100%	3.2	Visual-Manual
2	Proving Ground Pressure	1/		3.6	4.5.6
Special a.	Differential pressure and igniter delay	20		3.6	4.5.6
Major 101	Base pad not in true position with respect to outer diameter of charge (to be determined after straps are securely tied and charge is laced)		100%	3.2	Gage

NOTES:

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PARAGRAPH 4.4.2.23	TITLE CONTINUATION SHEET		SHEET 2 OF 3		DRAWING NUMBER 9291897
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
102	Dimensional gage failure (maximum length and diameter)		100%	3.5.1	Gage
103	Lacing missing, improper or insecure		100%	3.2	Visual-Manaul
104	Igniter protector cap missing or improperly assembled		0.40%	3.2	Visual
105	Propellng charge assembly weight		100%	3.4.2	4.5.8
106	External threads or chains of excessive length (including loose threads) clinging to the outside of bag		0.40%	3.2	Visual
107	Dimensional Gage Failure (Minimum Diameter)		100%	3.5.2	Gage

NOTES:
1/ Pressure sampling will consist of monitoring pressure on differential pressure and muzzle velocity standard deviation samples.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.23	CONTINUATION SHEET	3	3	9291897
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
108	Any component missing or improperly assembled (including flash reducer)		100%	3.2	Visual
109	Flash reducer (assembly) damaged to the extent that potassium sulfate can escape		100%	3.2	Visual
110	Strap loose, broken or insecurely tied (prior to assembly of lacing jacket to the charge)		0.40%	3.2	Visual-Manual
	Proving Ground				
111	Muzzle velocity and muzzle velocity standard deviation	15		3.6	4.5.6
Minor					
201	Location of lacing jacket		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48243D (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.24	Container, Prior to Insertion of Charge				9293303
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Foreign matter or rust in container		100%	3.2	Visual
102	Bottom cushion missing		100%	3.2	Visual
103	Charge not tightly wrapped immediately prior to insertion into container (i.e., all charge full diameter points shall be in contact with the wrapper)		0.40%	3.2	Visual
Minor					
201	Damaged container (to include dents protruding greater than 1/4" into container)		0.65%	3.2	Visual-Gage (see 6.4)
202	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER 9293303
4.4.2.25	Container, Prior to Sealing		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Wrapper or long filler above top of charge except for one allowable torn filler (to be inspected with charge in container)		100%	3.2	visual
102	Charge diametrically loose to the extent that a long filler may be inserted		100%	3.2	Manual
103	Any component missing or improper		100%	3.2	Visual
104	Dimension "X", minimum (in lieu of measuring Dimension "X", the measurement of Dimension "Y", maximum may be taken)		100%	3.2	Gage
105	Filler inserted in container prior to inserting top cushion		0.40%	3.2	Visual
106	More than five long fillers present		0.40%	3 .2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
4.4.2.25	CONTINUATION SHEET 2		2		9293303
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
107	Overlap of one long filler with another		0.40%	3.2	Visual
108	Charge cannot be removed by hand	1/		3.2	4.5.11
109	Top cushion incorrect type; i.e., cushion does not have hole		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		0.65%	3.11	Visual

NOTES:

1/ One sample from eight (8) hours production.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.26	Container, Sealed		9293303
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Test plug missing		100%	3.2	Visual
102	Air pressure test		100%	3.2	4.5.4
103	Test plug assembly torque		100%	3.7	Test
104	DOD symbol missing, misleading or unidentifiable (if applicable)		0.40%	3.2	Visual
Minor					
201	Test plug not greased (to be checked after air pressure test and prior to reassembly)		0.65%	3.2	Visual
202	Metallic seal missing or incorrectly applied		0.65%	3.2	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2	OF 2	DRAWING NUMBER 9293303
4.4.2.26	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
203	Marking missing, misleading, or unidentifiable		0.65%	3.2	Visual
204	Cover incompletely engaged		0.65%	3.2	Visual

NOTES:

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48243D (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9293303
4.4.2.27	Containerized Charge				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Note (a)	Tear down Inspection	(b)		(c) (d)	4.5.7

NOTES:

(a) Category is critical or major dependent upon the category of the defect inspected.
(b) See paragraph 4.4.3.1.
(c) Requirement paragraph is dependent upon inspection performed.
(d) Allowance will be made for stretching of cloth or relaxation of the charge when verifying characteristics that were established prior to loading at loading and assembly. Allowance shall not exceed twice original tolerance or one-half minimum tolerance.

NOTES:

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4.4.3 Testing

4.4.3.1 Teardown inspection

4.4.3.1.1 Initial production lot sampling. In addition to the samples of 4.4.3.1.2, three completely assembled charges shall be randomly selected from each day production until three consecutive days production is found acceptable.

4.4.3.1.2 Lot sampling. Thirty completely assembled charges shall be randomly selected from each lot. If there is a lapse in production in excess of ninety (90) days, or the previous lot was waived or rejected due to this inspection, sampling will return to initial production lot sampling.

4.4.3.2 Moisture content of black powder. - At least one sample of black powder from each eight hours loading of base igniter assemblies will be tested for moisture content.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 of this specification.

4.5 Methods of inspection.

4.5.1 Weight of black powder in base igniter loading assembly. - The inspections below are to be conducted by inspection personnel using equipment designated for inspection purposes and shall be clearly distinguished from production operations. The weight of the black powder for each compartment will be inspected 100 percent. The weight of the black powder in the basepad will be determined by subtracting an average weight of the empty assembly or the actual weight as marked on the individual asssembly from the loaded basepad and empty igniter core bag assembly weight. The weight of the black powder in the igniter core bag will be determined by subtracting the actual weight of the loaded basepad and empty igniter core bag assembly from the loaded base igniter loading assembly weight.

4.5.2 Propellant weight (see 3.4.) The inspections below are to be conducted by inspection personnel using equipment designated for inspection purposes and shall be clearly distinguished from production operations. An increment which fails to comply with the applicable requirements shall be classed defective and removed from the lot. Inspection equipment shall be in accordance with 4.4.4.

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4.5.2.1 Method 1. The loaded increments will be inspected 100 percent. The weight of the propellant will be determined by subtracting an average weight of the empty increment bag or the actual weight as marked on the individual increment bag from the loaded increment weight.

4.5.2.2 Method 2. The weight of the propellant will be inspected 100 percent on a different balance prior to filling the increment. In addition the propellant weight of the loaded increment will be inspected using a sampling plan from MIL-STD-1235 with an AQL of 0.40 percent. The weight of the propellant will be determined by subtracting an average weight of the empty increment bag or the actual weight as marked on the individual increment bag from the loaded increment weight.

4.5.3 Moisture content of black powder. The moisture content of the black powder will be determined in accordance with MIL-P-223. If any sample fails to meet the specified requirement the eight hours production represented by the sample shall be rejected.

4.5.4 Air pressure test. Each container shall be tested as specified on the appropriate drawing. A pressure drop equivalent to or greater than the applicable requirement shall be cause for rejection and removal from the lot.

4.5.5 Cloth and thread

4.5.5.1 Cloth, spun viscose rayon resin impregnated. Each roll, as a minimum, shall be tested for compliance with the physical characteristics listed in Table I in 3.5 of MIL-C-43157. Testing will be in accordance with 4.3.1 of MIL-C-43157. Any sample which fails to comply with the applicable requirement shall be cause for rejection of the cloth bolt represented by the sample. The method of inspection to be used during production is to be established by the contractor and approved by the government. The method, as a minimum, shall contain positive assurance that the material being utilized has been verified for compliance with the specification requirements as stated above. (see 6.12)

4.5.5.2 Cloth. At the time cloth is introduced to the sewing or cutting operation, an identification will be made for each roll to verify that proper material is used. Any cloth failing to be identified as proper material in accordance with applicable drawings and specifications shall be rejected from the lot.

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4.5.6 Proving ground (See 3.6). - This test shall be conducted at a Government Proving Ground in accordance with the applicable acceptance test procedure using M549, RA or PXR-6325 projectiles inert loaded within 0.1 pound of an as fired weight of 96.0 pounds using primer, M82 and fired in cannon M199 with 75 percent (minimum) remaining life. Firing shall be conducted with a spindle to charge base stand-off of 1 inch (plus or minus 1/4 inch). Prior to firing, all charges shall be temperature conditioned in a large conditioning box with adequate air circulation at the required temperature for a minimum of 24 hours and assurance will be made that, when fired, the charges are at the conditioning temperature. The following tolerances shall apply to all -65°F, 145°F and 70°F conditioning: -65°F from -60°F to -65°F; 145°F from 140°F to 145°F; 70°F from 68°F to 72°F. Copper crusher gages (2) shall be used to measure the maximum chamber pressure of each charge. Temperature indicator (thermometer) strips shall be placed on the side (lacing jacket) of all charges or on the side of the metal containers, as applicable, conditioned at 145°F as a further assurance of the proper conditioning temperature. If any indicator strip shows the charge has been exposed to a temperature greater than 145°F, that charge will not be fired.

4.5.6.1 Muzzle velocity and muzzle velocity standard deviation. Determination of acceptance will be in accordance with MIL-STD-414, Section B, Table B3, AQL of 0.65%. All muzzle velocity and muzzle velocity standard deviation samples will be ballistic tested at 70°F. Fifteen (15) charges will be fired for muzzle velocity and muzzle velocity standard deviation. If the sample is determined unacceptable the lot will be rejected. In addition, if the velocity standard deviation of the sample multiplied by the factor 0.76 exceeds the applicable lot requirements, the lot shall be rejected. Fifteen (15) calibration charges will be fired alternately with the test samples and the muzzle velocity and pressure of all test charges will be corrected to standard conditions.

4.5.6.2 Differential pressure. Two pressure-time gages at different locations (one in the spindle and one at the projectile base) will be used to record pressure-time traces. Differential pressure traces will be obtained by subtracting the projectile gage output from the spindle gage output. The maximum differential pressure of any charge will be defined as maximum amplitude of the first negative wave of the differential pressure trace prior to peak pressure. Any trace which contains no amplitude in the negative direction will be considered to have a differential pressure of zero. Any sample

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failing to meet the maximum differential requirements shall be cause for the lot to be referred (see 3.6.1). Ignition delay will be recorded on all differential pressure firings. Any sample failing to meet the ignition delay requirements shall be cause for the lot to be referred. Ten (10) charges will be conditioned to 145°F and ten (10) charges will be conditioned to -65°F for differential pressure firings. Ten (10) calibration charges will be fired intermixed (five (5) at each temperature) with the test samples and the muzzle velocity and pressure of all test charges corrected to standard conditions.

4.5.6.3 Pressure. - If any propelling charge individual corrected maximum pressure exceeds the applicable requirement, the lot shall be rejected.

4.5.7 Teardown inspection. - Each container and propelling charge will be subjected to each of the critical (except critical defect 2, paragraph 4.4.2.23 and 100 percent major (except major defect 101, paragraph 4.4.2.9; major defect 102, paragraph 4.4.2.11; major defect 101, paragraph 4.4.2.21; major defect 101, major defect 105, paragraph 4.4.2.23; major defect 104, paragraph 4.4.2.25) inspections defined in 4.4.2 of this specification. Samples may be accummulated or inspected upon selection providing the selection remains random and represents the lot. The inspections will be performed by the contractor and witnessed by the Government. The detection of any one of the above defects shall be cause for rejection of the lot represented by the sample.

4.5.8 Charge, propelling, assembly weight. The actual weights as marked on the individual complete igniter assembly, and empty increment assembly and the actual or average weight of the flash reducer, plus the averge weight of the lacing cords and jacket, together with the nominal propellant weights as specified in the loading authorization shall be subtracted from the assembled charge, propelling, assembly weight. If the resulting weight deviation is greater than the requirement, the propelling charge will be classified defective and removed from the lot (see 6.9).

4.5.9 Complete igniter assembly weight.

4.5.9.1 Method 1. The actual weight of the tube and averge weight of the cap along with the average weight of the

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empty base igniter assembly or the actual weight as marked on the individual empty base igniter assembly plus the average weight of the body and strap assembly or the actual weight of the body and strap assembly shall be recorded for each complete igniter assembly. These weights along with the nominal weights of black powder shall be subtracted from the complete igniter assembly weight. If the resulting weight deviation is greater than the requirement, the complete igniter assembly will be classified defective and removed from the lot (see 6.10.1).

4.5.9.2 Method 2. The average weight of the empty base igniter assembly of the actual weight as marked on the individual empty base igniter assembly plus the average weight of the body and strap assembly or the actual weight of the body and strap assembly shall be recorded for each complete igniter assembly. This weight along with the average weight of the tube and cap and the nominal weights of black powder shall be subtracted from the complete igniter assembly weight. If the resulting weight deviation is greater than the requirement, the complete igniter assembly will be classified defective and removed from the lot (see 6.10.2).

4.5.10 Complete igniter alignment. Each complete igniter assembly shall be assembled to the increment assembly utilizing an approved method of assuring alignment between the igniter tube and the outer diameter of the increment assembly. The method to be used during production is to be established by the contractor and approved by the Government (see 6.13).

4.5.11 Charge removal from container. Each sample shall be tested for removal of the propelling charge from the container by hand. The charge may be pulled out by the tying straps or the container may be inverted and the charge either pulled out or slid out. Each charge tested shall be checked for damage to the tying straps (looseness, tears, etc.) and the charge will be repaired (retie the tying straps and replace the charge, if necessary). Each charge tested shall be rewrapped (may use the same wrapper if the wrapper is not badly damaged) and re-packed in accordance with this specification and the applicable drawings. Any charge that cannot be removed from the container using the above method shall be cause for rejection of the production represented by the sample.

5. PACKAGING

5.1 Packing.

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5.1.1 Level A - The propelling charges shall be packed, sealed and tested in accordance with the specified requirements of drawing 9293303.

5.1.2 Level B - Same as level A.

5.1.3 Level C - Same as level A.

5.2 Marking.

5.2.1 Level A - The propelling charges shall be marked in accordance with the specified requirements of drawing 9293303.

5.2.2 Level B - Same as Level A.

5.2.3 Level C - Same as level A.

6. NOTES

6.1 Intended use. - The propelling charges covered by this specification are intended for use in the 155MM Howitzer, M198.

6.2 Ordering data. - See MIL-A-48078.

6.3 Submission of designs for approval. - See 6.2.3 of MIL-A-48078. - Submit equipment designs, as required, to Commander, US Army Armament Research and Development Center, Attn: AMSMC-QAR-I (D), Dover, New Jersey 07801-5001.

6.4 Visual examination qualification. - When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

6.5 Inspection lot size. - It is noted that the size of inspection lots of components, assemblies, or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all the lot formation criteria and sampling provisions of this specification are maintained.

6.6 Proving ground. - A copy of all proving ground data should be submitted to Commander, US Army Armament Research and Development Center, Attn: AMSMC-QAR-R (D) and SMCAR-AEE-BP, Dover, New Jersey 07801-5001.

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6.7 Teardown inspection. - A list of all defects found during teardown inspection shall be submitted to Commander, US Army Armament Research and Development Center, Attn: AMSMC-QAR-R (D), Dover, New Jersey 07801-5001.

6.8 Charge, propelling assembly weight and complete igniter assembly weight. - This requirement applies to the load plant at the time of loading only. Due to subsequent change in tare weight of the components because of variation in material and moisture content, this test cannot be repeated after initial manufacture to the degree of accuracy required.

6.9 Charge, propelling, assembly weight. -

Weights To Be Determined	Assembly Drawing Number
Actual Weight of Complete Igniter Assembly	9281940
Average/Actual Weight of Flash Reducer Loading Assembly	9289200
Actual Weight of Body and End Assembly	9278967
Average Weight of Lacing Cord	9281897
Average Weight of Lacing Jacket	9281895
Nominal Propellant Weight	Loading Authorization
Component Weight (sum of above weights)	

The propelling charge assembly weight minus the component weight equals the "weight deviation".

6.10 Complete igniter assembly weight

6.10.1 Method 1. -

Weights To Be Determined	Assembly Drawing Number
Actual Weight of Tube	9285215
Average/Actual Weight of Base Igniter Assembly	9278985

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Average/Actual Weight of Body and Strap Assembly	9278984
Average Weight of Cap or Tube Closure	9289199/ 9345159
Nominal Weights of Black Powder Component Weight (sum of above weights)	9281936

The complete igniter assembly weight minus the component weight equals the "weight deviation".

6.10.2 Method 2. -

Weights To Be Determined	Assembly Drawing Number
Nominal Weight of Tube	See MIL-I-48252
Average/"Actual Weight of Base Igniter Assembly	9278985
Average/Actual Weight of Body and Strap Assembly	9278984
Average Weight of Cap or Tube Closure	9289199/ 9345159
Nominal Weights of Black Powder Component Weight (sum of above weights)	9281936

The complete igniter assembly weight minus the component weight equals the "weight deviation".

6.11 First article inspection procedure. - Sample propelling charges shall be submitted to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QAR-R (D), Dover, New Jersey 07801-5001, upon implementation of this military specification of those items procured for the U. S. Army or otherwise specified by the contracting officer.

6.12 Method of inspection submission. - Methods of inspection for resin impregnated spun viscose rayone cloth shall be submitted to Commander, US Army Armament Research and Development Center, Attn: AMSMC-QAR-R(D), Dover, New Jersey 07801-5001 for approval.

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6.13 Method of assuring alignment submission. - The method for assuring alignment of the complete igniter assembly shall be submitted to Commander, US Army Armament Research and Development Center, Attn: AMSMC-QAR-R(D), Dover, New Jersey 07801-5001 for approval.

6.14 Differential pressure and ignition delay. - The values given in this specification do not represent a definite boundary between safe and unsafe conditions but serve as triggers to require a safety analysis of the data. If the differential pressure or ignition delay requires referral of the lot, the disposition of the lot shall be referred to US Army Armament Research and Development Center, Attn: AMSMC-QAR-R (D), Dover, New Jersey 07801-5001.

6.15 Proving ground summary (see 4.5.6).

Temp	Sample Size	Number of Ref Rounds	Velocity	Maximum Individual Pressure	Differential Pressure
70°F	15	15	*	50,000	NA
145°F	10	5	INFO	57,900	**
-65°F	10	5	INFO	NA	**

* See 4.5.6.1 and 3.6
** See 3.6

6.16 Individual maximum pressure. - Individual Maximum Pressure at 145°F is based on the PIMP pressure (heaviest projectile weight zone) minus 850 psi, the effect of projectile weight on pressure since the standard projectile weight zone is being tested.

6.17 Data cards. - A copy of all data cards should be submitted to Commander, US Army Armament Research and Development Center, Attn: AMSMC-QAR-R (D) and SMCAR-AEE-BP, Dover, New Jersey 07801-5001.

6.18 Assembly torque. - The assembly torque of the test plug will apply at time of loading only due to the relaxation of the gasket material.

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6.19 International standardization agreements. - Certain provisions (see 3.6 and 6.15) of this specification are the subject of international standardization agreement (Quadrilateral MOU Concerning the Standardization of Elements of 155MM Howitzer and Ammunition, 28 April 78). When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconcilliation action through Project Manger, Cannon Artillery Weapons Systems/Joint Project Manger, Semi-Active Laser Guided Projectiles (PM-CAWS/JPM SAL GP).

6.20 Temperature indicator strips. The highest temperature reading from each indicator strip shall be recorded for informational purposes and reported on the applicable firing record.

6.21 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Reasearch and Development Center (ARDC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, ARRADCOM or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.22 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
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Preparing activity
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-48243D	CHARGE PROPELLING, 155MM, M203, LOADING, ASSEMBLING AND PACKING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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